

Nightmares Course: Anaphylaxis on the Ward

Section 1: Case Summary

Scenario Title:	Anaphylaxis on the Ward
Keywords:	Anaphylaxis, shock, medication error
Brief Description of Case:	This case involves the identification and initial management of anaphylaxis. The patient will become progressively more hypotensive and develop tongue swelling if not given IM epinephrine (0.3mg-0.5mg every 5mins). If given appropriate repeated epi doses, then the patients' hemodynamics will improve and symptoms will partially resolve. Adjunctive meds (antihistamines and steroids) may be given but will not affect symptoms acutely.

Goals and Objectives	
Educational Goal:	To practice the initial assessment and management of anaphylaxis
Objectives: (Medical and CRM)	By the end of this simulation exercise, all learners will be able to: 1. Recognize clinical signs/symptoms of the patient in anaphylaxis 2. Demonstrate appropriate initial management of anaphylaxis 3. Review the pathophysiology of anaphylaxis and initial treatment priorities 4. Compare and contrast how epinephrine is packaged on the crash cart (1:10,000) vs ampule (1:1000) 5. Appropriately ask for help and communicate concerns over the phone (i.e. to RACE team or attending) 6. Review communication strategies in a. Disclosing medical errors to patients b. Situation development of patient's clinical status
EPAs Assessed:	

Learners, Setting and Personnel	
Target Learners:	☒ Junior Learners ☐ Senior Learners ☐ Staff
	☒ Physicians ☐ Nurses ☐ RTs ☐ Inter-professional
	☐ Other Learners:
Location:	☒ Sim Lab ☐ In Situ ☐ Other:
Recommended Number of Facilitators:	Instructors: 1
	Sim Actors: 1
	Sim Techs: 1

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
Affiliations/Institutions(s):	Queen's University
Contact E-mail:	chaplintim2@gmail.com
Last Revision Date:	Jan 2020
Revised By:	Dr. Chris Heyd
Version Number:	1



Nightmares Course: Anaphylaxis on the Ward

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Keith Anderson		Age: 68		Gender: M	Weight: 80kg
Presenting complaint: Rash					
Temp: 36.5 °C	HR: 122	BP: 86/40	RR: 18	O ₂ Sat: 96%	FiO ₂ : Room air
Cap glucose: 5.6			GCS: 15		
Triage note: It's 9:00 pm and you have been called to assess a patient who has developed a rash and nausea shortly after starting an IV dose of pip/tazo. On handover you were told the patient was a 68M who was admitted this morning for treatment of a diabetic foot infection that has been progressing despite oral antibiotics.					
Allergies: Penicillin (Unknown reaction)					
Past Medical History: Type 2 diabetes			Current Medications: Metformin PO 1g BID Sitagliptin PO 100mg daily Canagliflozin PO 200mg daily		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
RN to provide when asked: You've taken care of Mr. Anderson since the beginning of your shift 2 hours ago. He is a very pleasant gentleman admitted for a severe diabetic foot infection who developed a whole body rash, throat itchiness, and nausea about 5 minutes after you started a pip/taz infusion. You call the resident for help immediately.	
IF asked, you can consult the patient's chart and determine that he is only on piperacillin-tazobactam, metformin and oral morphine prn. He is full code. He is allergic to penicillin (this was apparently unknown to the admitting team during the day – they wrote the orders for the pip-taz)	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Nil	Neuro: Nil
Resp: Diffuse wheeze	Head & Neck: Nil
Abdo: Soft, non-tender	MSK/skin: Diffuse urticarial rash on trunk and arms
Other: No visible oral swelling	

Nightmares Course: Anaphylaxis on the Ward

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin Adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
None
C. Required Medications
Cardiac epinephrine Epinephrine ampule Epi-pen Ventolin
D. Moulage
Hospital gown Diffuse urticaria rash 20G IV with medication infusing
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i>
Diffuse wheeze bilaterally

Nightmares Course: Anaphylaxis on the Ward

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside RN	RN NOTES: - You've taken care of Mr. Anderson since the beginning of your shift 2 hours ago. He is a very pleasant gentleman admitted for a severe diabetic foot infection who developed a whole body rash, throat itchiness, and nausea about 5 minutes after you started a pip/taz infusion. You call the resident for help immediately. - IF asked, you can consult the patient's chart and determine that he is only on piperacillin-tazobactam, metformin and oral morphine prn. He is full code. He is allergic to penicillin (this was apparently unknown to the admitting team during the day – they wrote the orders for the pip-taz)

Nightmares Course: Anaphylaxis on the Ward

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tach HR: 122 BP: 86/40 RR: 18 O ₂ SAT: 96% T: 36.5°C GCS: 15	<i>Pt: "I'm feeling a bit short of breath and light headed"</i>	<u>Expected Learner Actions</u> ☐ Monitors ☐ Focused history and physical ☐ Place patient on oxygen ☐ Stop antibiotic infusion ☐ Order IM epinephrine (0.3-0.5mg) ☐ Call for help	<u>Modifiers</u> <u>Triggers</u> <i>For progression to next state</i> - No Epi or ongoing infusion at 3 min → 2. Worsening - All actions complete → 3. Improving	
2. Worsening Rhythm: Sinus tach HR: 140 BP: 75/35 RR: 24 O ₂ SAT: 96%	<i>RN: "It looks like his lips are starting to swell. Do you think it could be the antibiotic?"</i>	<u>Expected Learner Actions</u> ☐ Stop antibiotic infusion ☐ Give IM epinephrine 0.3-0.5mg	<u>Modifiers</u> - If no epi IM given, RN to prompt <u>Triggers</u> - All actions complete → 3. Improving	
3. Improving Rhythm: Sinus tach HR: 122 BP: 100/40 RR: 18 O ₂ SAT: 96%	<i>Pt "I am feeling a little better but my throat is still itchy."</i> Normal lips and oropharynx on examination	<u>Expected Learner Actions</u> ☐ Reassess patient ☐ Ventolin MDI ☐ Steroid IV (e.g. methylprednisolone, dexamethasone, hydrocortisone) ☐ Antihistamine IV or PO ☐ Bolus of 1L normal saline ☐ Consider 2 nd dose IM Epi	<u>Modifiers</u> <u>Triggers</u> -All actions complete → 4. Stabilization	
4. Stabilization Rhythm: Sinus tach HR: 110 BP: 110/65 RR: 18 O ₂ SAT: 99%		<u>Expected Learner Actions</u> ☐ Discuss med error with patient ☐ Discuss disposition with RACE/Senior resident	<u>Modifiers</u> <u>Triggers</u> - All actions complete → END CASE	



Nightmares Course: Anaphylaxis on the Ward

Appendix A: Laboratory Results (admission bloodwork)

CBC

WBC 14
Hgb 130
Plt 220

Lytes

Na 137
K 3.8
Cl 97
HCO₃ 24
Urea 7.4
Cr 137

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

No imaging given in this case

Nightmares Course: Anaphylaxis on the Ward

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. Clear communication (with RN, staff on the phone, patient) and management priorities (SBAR)

S • B • A • R

Before you call, be prepared! Be clear, concise, focus on the problem & only report what is relevant to the current situation!

Be sure you do the following:

- Assess the patient.
- Review appropriate parts of the medical record (eg, flow sheet, MAR, physician notes/orders, labs).
- Determine the appropriate person to call.
- Have the medical record available when you call.
- Use the following form to organize your conversation.

Situation: 5-10 second “punch line” – What is happening now? What are the chief complaints or acute changes?

This is _____. *I’m calling about* _____

Background: What factors led up to this event? Pertinent history (eg, admitting diagnosis) & objective data (eg, vital signs, labs) that support how patient got here.

The patient has _____

Assessment: What do you see? What do you think is going on? A diagnosis is not necessary; include the severity of the problem.

I think the problem is _____

Recommendation: What action do you propose? State what the patient needs (get a time frame).

I request that you _____

Nightmares Course: Anaphylaxis on the Ward

2. Discuss the initial approach to the patient with anaphylaxis and its management algorithm
3. Understand available epinephrine concentration and doses (ie crash cart 1:10,000 vs ampule 1:1000)
4. Definition of anaphylaxis

Table 1. Definition of anaphylaxis. Clinical Criteria for Diagnosis

Anaphylaxis is highly likely when any one of the following 3 criteria are fulfilled:

1. Acute onset of an illness (minutes to several hours) with involvement of the skin, mucosal tissue, or both (eg, generalized hives, pruritus or flushing, swollen lips-tongue-uvula).
AND AT LEAST ONE OF THE FOLLOWING
 - a. Respiratory compromise (eg, dyspnoea, wheeze-bronchospasm, stridor, reduced PEF, hypoxaemia).
 - b. Reduced BP or associated symptoms of end-organ dysfunction (eg, hypotonia [collapse], syncope, incontinence).
2. Two or more of the following that occur rapidly after exposure to a likely allergen for that patient (minutes to several hours):
 - a. Involvement of the skin-mucosal tissue (eg, generalized hives, itch-flush, swollen lips-tongue-uvula).
 - b. Respiratory compromise (eg, dyspnoea, wheeze-bronchospasm, stridor, reduced PEF, hypoxaemia).
 - c. Reduced BP or associated symptoms (eg, hypotonia [collapse], syncope, incontinence).
 - d. Persistent gastrointestinal symptoms (eg, crampy abdominal pain, vomiting).
3. Reduced BP after exposure to known allergen for that patient (minutes to several hours):
 - a. Infants and children: low systolic BP (age specific) or greater than 30% decrease in systolic BP*.
 - b. Adults: systolic BP of less than 90 mmHg or greater than 30% decrease from that person's baseline.

PEF, Peak expiratory flow; BP, blood pressure. *Low systolic blood pressure for children is defined as less than 70 mmHg from 1 month to 1 year; less than (70 mmHg + [2 x age]) from 1 to 10 years; and less than 90 mmHg from 11 to 17 years. Reproduced with permission from Journal of Allergy and Clinical Immunology².

5. Communication method on breaking medication error
 - Begin by stating there has been an error;
 - Describe the course of events, using nontechnical language;
 - State the nature of the mistake, consequences, and corrective action;
 - Express personal regret and apologize;
 - Elicit questions or concerns and address them; and
 - Plan the next step and next contact with the patient.

References

1. <https://en.wikipedia.org/wiki/SBAR>
2. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6838992/>
- 3.

